

Week 1 Worksheet 2 Solutions

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Exercise 1. Translation. Let $\psi(x)$ be the position space wavefunction of a particle, and let $\varphi(p)$ be its wavefunction in momentum space.

- Find the momentum space wavefunction of the position wavefunction $\psi(x + a)$ for some constant a . Make sure your answer is in terms of $\varphi(p)$.
- Write down the Taylor series expansion of $\psi(x + a)$ about $a = 0$, and show it's the same as

$$e^{ipa/\hbar}\psi(x).$$

Remark. We say that $e^{-ipa/\hbar}$ is the **translation operator** (by a).

- We do the Fourier transform to find

$$\frac{1}{\sqrt{2\pi\hbar}} \int e^{-ipx/\hbar} \psi(x + a) dx = \frac{1}{\sqrt{2\pi\hbar}} \int e^{-ip(x-a)} \psi(x) dx = e^{ipa/\hbar} \varphi(p).$$

- This is easiest if you imagine that the function $\psi(x + a) = f_x(a)$ is a function of a at fixed x . We know how to expand these about $a = 0$: The Taylor series expansion about $a = 0$ is by definition

$$\begin{aligned} f_x(a) &= \sum_{n=0}^{\infty} \frac{f_x^{(n)}(0)}{n!} a^n \\ &= \sum_{n=0}^{\infty} \frac{a^n}{n!} \frac{\partial^n \psi(x)}{\partial x^n}. \end{aligned}$$

On the other hand, expanding

$$e^{ipa/\hbar} \psi(x) = \exp(a \partial_x) \psi(x)$$

as a power series, we find

$$\sum_{n=0}^{\infty} \frac{a^n}{n!} \partial_x^n \psi(x),$$

which exactly matches the expansion for $\psi(x + a)$.

Exercise 2. Probability Current. In class, you saw the quantity

$$J(x, t) = \frac{1}{2m}[\Psi^*(x, t)p\Psi(x, t) - \Psi(x, t)p\Psi^*(x, t)],$$

which is called **the probability current**.

- a) Show, using the time-dependent Schrödinger equation, that

$$\frac{\partial J}{\partial x} + \frac{\partial \rho}{\partial t} = 0,$$

where $\rho(x, t) = |\Psi(x, t)|^2$ is the probability density. This equation is called the **continuity equation**.

- b) Comment physically on what the continuity equation implies.

Hints: The 3D analog of the continuity equation is $\nabla \cdot \mathbf{J} + \partial \rho / \partial t = 0$. If you have a charged particle of charge q , and you multiply the continuity equation by q , what do you get?

- a) We first evaluate

$$\begin{aligned} \partial_t |\Psi|^2 &= \partial_t \Psi^* \Psi + \Psi^* \partial_t \Psi = \left(-i \frac{\hbar}{2m} \partial_x^2 \Psi^* + \frac{i}{\hbar} V \Psi^* \right) \Psi + \Psi^* \left(i \frac{\hbar}{2m} \partial_x^2 \Psi - \frac{i}{\hbar} V \Psi \right) = \\ &= i \frac{\hbar}{2m} (\Psi^* \partial_x^2 \Psi - \Psi \partial_x^2 \Psi^*). \end{aligned}$$

On the other hand,

$$\partial_x J = -i \frac{\hbar}{2m} [\Psi^* \partial_x^2 \Psi - \Psi \partial_x^2 \Psi^*],$$

where the terms of the form $|\partial_x \Psi|^2$ cancel. Once we have $\partial_t \rho$ and $\partial_x J$ in the above forms, we see that they exactly cancel.

- b) By analogy with QM, we see that if we multiply the continuity equation by q we get the continuity equation for electromagnetism, which says that charge cannot be created or destroyed. So in QM the continuity equation must say that *probability* cannot be created or destroyed.